How Mean Fields Saturate the Cosmic-Ray Non-Resonant Streaming Instability

Mean-Field Theory & MHD-PIC Simulation

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Outline

1. **Background & Motivation**
2. **Mean-Field Theory for the Fixed CR-Current Case**
   - 2.1 Analytical Theory
   - 2.2 Multi-dimensional MHD Simulations
3. **Mean-Field Theory for the Self-Consistent CR Case**
   - 3.1 Analytical Theory
   - 3.2 One-dimensional MHD-PIC Simulations
4. **Conclusion, Discussion & Extension**

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2.1 Analytical Theory
2.2 Multi-dimensional MHD Simulations

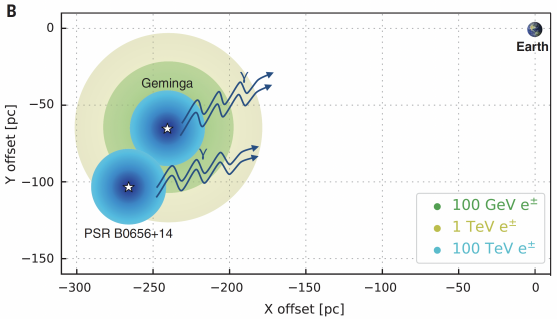
3. Mean-Field Theory for the Self-Consistent CR Case
3.1 Analytical Theory
3.2 One-dimensional MHD-PIC Simulations

4. Conclusion, Discussion & Extension

Pulsar Wind Nebulae

- Middle-aged pulsars are local accelerators of leptons (Geminga and PSR B0656+14)
  - suppressed particle diffusion inferred from TeV γ-ray halo
  - magnetic-field amplification inferred from X-ray emission

  $\Longrightarrow$ a potential explanation: the CR transport is affected by the self-generated enhanced turbulence, in which case the resonant streaming instability doesn't seem to be effective enough [R.López-Coto et al. 2022 NA]

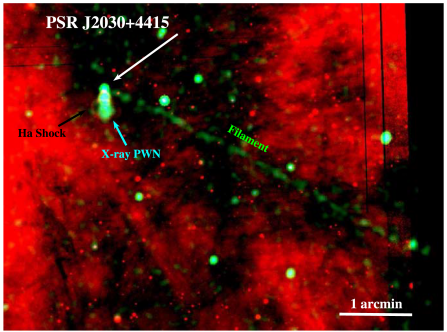


Schematic illustration of HAWC detection [A.U.Abeysekara et al. 2017 Science]

Bow Shock PWNe

- X-ray non-thermal filamentary emissions due to high-energy leptons (BSPWN PRS J2030+4415)
  - filament magnetic field $\sim 100\mu G$ inferred from X-ray emission, possibly generated by injected particles

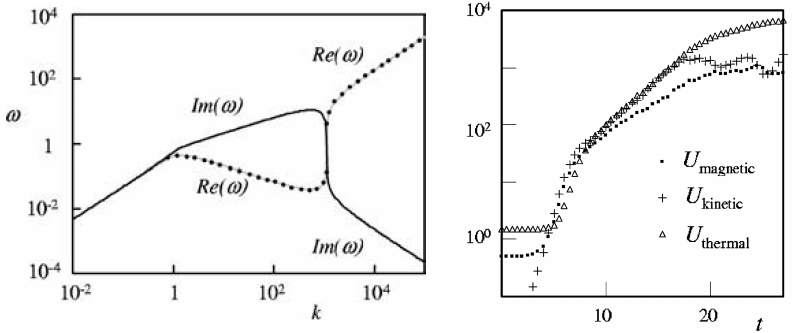
  $\Longrightarrow$ a similar possible physical picture: the lack of scattering allows the current-carrying particles to stream ballistically along the filament and excite magnetic fields [B.Olmi et al. 2024 AA]



Morphology of bow shock, PWN, and filament [M.de Vries & R.W.Romani 2020 ApJL]

NRSI & MHD Simulations

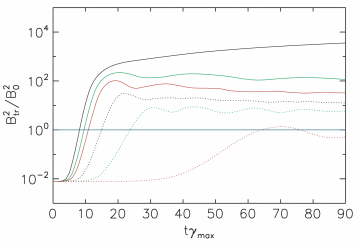
- Bell investigated a non-resonant, nearly purely growing modes which leads to strong magnetic-field amplification
  - a fixed CR-current $J_{cr} = en_{cr}u_{cr}$ in the direction of $\mathbf{B}_0$ drives the non-resonant streaming instability (NRSI) characterized by $k_{\max} = 2\pi J_{cr}/(cB_0)$, $\gamma_{\max} = k_{\max} v_A$, and $|\omega_{kr}| \ll \gamma_{\max}$

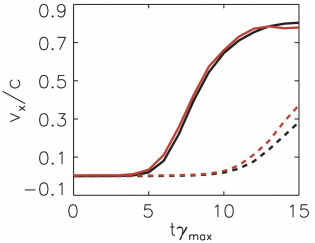


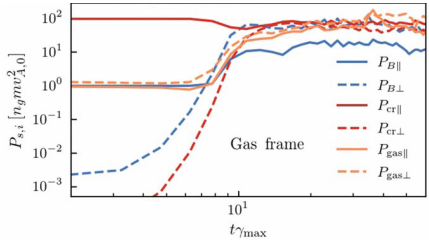
Linear dispersion relation & magnetic-field amplification in MHD simulation [A.R.Bell 2004 MNRAS]

Kinetic Simulations

- Kinetic simulations of NRSI saturation [M.A.Riquelme & A.Spitkovsky 2009 ApJ; G.Zacharegkas et al. 2024 ApJ]
  - wave dynamics: damping on the short wavelength modes & growth of the large wavelength modes
  - mean-field dynamics: the acceleration of the background plasma in the direction parallel to $\mathbf{B}_0$
  - particle dynamics: CR scattering, diffusion and isotropization







Magnetic-field amplification, bulk plasma acceleration & CR anisotropic pressure evolution in kinetic simulations

Outline

- A quantitative study delivered here to figure out the leading-order physics, the corresponding minimal model capturing the dynamics, and the saturation criteria:
  - the fixed CR-current case $\mathbf{J}_{cr} = en_{cr}u_{cr}\hat{\mathbf{b}}_0$
  - the self-consistent CR case $\bar{\mathbf{J}}_{cr\parallel}(t) = en_{cr}\bar{u}_{cr\parallel}(t)\hat{\mathbf{b}}_0$ given an initial CR beam $\mathbf{J}_{cr\parallel 0} = en_{cr}u_{cr\parallel 0}\hat{\mathbf{b}}_0$
- We consider a periodic homogeneous system with initial random-noise fluctuations, such that the energy and the momentum are globally conserved
- We focus on the NRSI saturation via its self-generated mean field dynamics, especially the bulk plasma motion (parallel mean flow) and the bulk CR motion

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MHD-PIC Model

- Model equations: ideal MHD + non-relativistic particles

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$d(p\rho^{-\gamma})/dt = 0$$

$$\rho(\partial_t + \mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + (\nabla \times \mathbf{B}) \times \mathbf{B}/(4\pi) - \mathbf{F}_{cr}$$

$$\partial_t \mathbf{B} = \mathbf{B} \cdot \nabla \mathbf{u} - \mathbf{u} \cdot \nabla \mathbf{B}$$

  - with CR force term $\mathbf{F}_{cr} = (\mathbf{J}_{cr} - en_{cr}\mathbf{u}) \times \mathbf{B}/c$ and CR current density $\mathbf{J}_{cr} = en_{cr}u_{cr}\hat{\mathbf{b}}_0$
  - $en_{cr}\mathbf{u} \times \mathbf{B}/c$ arises due to the quasi-neutrality condition of the composite plasma and CRs
- Conservative form of MHD-PIC model used by PLUTO [A.Mignone 2018 ApJ]

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot [(p + B^2/8\pi)\mathbf{I} + (\rho \mathbf{u}\mathbf{u} - \mathbf{B}\mathbf{B}/4\pi)] = -\mathbf{F}_{cr}$$

$$\partial_t E + \nabla \cdot [(E + P^*)\mathbf{u} - (\mathbf{u} \cdot \mathbf{B})\mathbf{B}/4\pi] = -\mathbf{u} \cdot \mathbf{F}_{cr}$$

$$E = \frac{\rho u^2}{2} + \frac{B^2}{8\pi} + \frac{p}{\gamma - 1}, \quad P^* = p + \frac{B^2}{8\pi}$$

Mean-Field Theory

- Mean-field assumptions:
  - direct wave-wave interactions among fluctuations ignored
  - the parallel mean flow $\bar{u}_\parallel$ is the effective mean-field interacting with fluctuations
- We consider nonlinear interactions between fluctuations $(\delta\mathbf{u}, \delta\mathbf{B}, \delta\rho, \delta p)$ and the parallel mean flow $\bar{u}_\parallel$ with variables $\delta\mathbf{u} = \delta u_\parallel \hat{\mathbf{b}}_0 + \delta\mathbf{u}_\perp$, $\delta\mathbf{B} = \delta B_\parallel \hat{\mathbf{b}}_0 + \delta\mathbf{B}_\perp$
  - Fourier decomposition e.g. $\delta\mathbf{B}_\perp = \sum_k \delta\mathbf{B}_{k\perp} \exp(i\mathbf{k}\cdot\mathbf{r} - i\omega_k t)$, with $\mathbf{k} = k_\parallel \hat{\mathbf{b}}_0 + \mathbf{k}_\perp$ and $\omega_k = \omega_{kr} + i\gamma_k$
- Nonlinear fluctuating-field equations for the $\mathbf{k}$-mode obtained from the MHD-PIC model:

$$
\begin{aligned}
-i(\omega_k - k_\parallel \bar{u}_\parallel)\delta\rho_k + i\rho_0 \mathbf{k}\cdot\delta\mathbf{u}_k &= 0 \\
-i(\omega_k - k_\parallel \bar{u}_\parallel)\delta p_k + i\gamma p_0 \mathbf{k}\cdot\delta\mathbf{u}_k &= 0 \\
-i\rho_0(\omega_k - k_\parallel \bar{u}_\parallel)\delta\mathbf{u}_k + i\mathbf{k}\delta p_k - i(\mathbf{k}\times\delta\mathbf{B}_k)\times\mathbf{B}_0/(4\pi) + \mathbf{J}_{cr}\times\delta\mathbf{B}_k/c &= \\
-q_{cr} n_{cr}\delta\mathbf{u}_k \times \mathbf{B}_0/c - e n_{cr}\bar{u}_\parallel \hat{\mathbf{b}}_0 \times \delta\mathbf{B}_k/c &= 0 \\
-i(\omega_k - k_\parallel \bar{u}_\parallel)\delta\mathbf{B}_k - ik_\parallel B_0 \delta\mathbf{u}_k &= 0
\end{aligned}
$$

  - $\omega_k - k_\parallel \bar{u}_\parallel$ originates from the total derivative $d/dt \equiv \partial_t + \mathbf{u}\cdot\nabla = \partial_t + \bar{u}_\parallel \nabla_\parallel$
  - $e n_{cr}\bar{u}_\parallel \hat{\mathbf{b}}_0 \times \delta\mathbf{B}_k/c$ originates from $q_{cr} n_{cr}\mathbf{u}\times\mathbf{B}/c$

Nonlinear Dispersion Relation

- Solving the fluctuating-field equation gives the 3D nonlinear dispersion relation (NLDR):

$$\left[1-\frac{k_\parallel^2 v_A^2}{(\omega_k-k_\parallel\bar{u}_\parallel)^2}\right]^2\left[1-\frac{k^2c_s^2}{(\omega_k-k_\parallel\bar{u}_\parallel)^2}\right]-\left(\frac{2\epsilon\gamma_{\max}}{\omega_k-k_\parallel\bar{u}_\parallel}\right)^2\left[1-\frac{k_\parallel(u_{cr}-\bar{u}_\parallel)}{\omega_k-k_\parallel\bar{u}_\parallel}\right]^2\left[1-\frac{k_\parallel^2c_s^2}{(\omega_k-k_\parallel\bar{u}_\parallel)^2}\right]=0$$

  with $k_{\max}=2\pi J_{cr}/(cB_0)$, $\gamma_{\max}=k_{\max}v_A$, $\epsilon=v_A/u_{cr}$, v_A the Alfvén speed, and c_s the sound speed
- 3D NLDR reduces to the 1D case:

$$(\omega_k-k_\parallel\bar{u}_\parallel)^2-k_\parallel^2c_s^2=0$$

$$\underbrace{(\omega_k-k_\parallel\bar{u}_\parallel)^2-k_\parallel^2v_A^2}_{\text{Alfvén wave}}=\underbrace{\mp\,2\epsilon\gamma_{\max}(\omega_k-k_\parallel u_{cr})}_{\text{CR-current drive}}$$

- Real frequency and growth rate of the non-resonant mode:

$$\omega_{kr}=\epsilon\gamma_{\max}+k_\parallel\bar{u}_\parallel \qquad \text{(upward frequency-sweeping)} \tag{1}$$

$$\gamma_k=\sqrt{2k_\parallel v_A\gamma_{\max}(1-\bar{u}_\parallel/u_{cr})-k_\parallel^2v_A^2-\epsilon^2\gamma_{\max}^2} \qquad \text{(saturation)} \tag{2}$$

  - $(\omega_k-k_\parallel\bar{u}_\parallel)$ is the Doppler-shifted frequency due to the PMF $\bar{u}_\parallel$ effect
  - $(u_{cr}-\bar{u}_\parallel)$ suggests that the PMF reduces the growth rate of NRSI $\Longrightarrow$ saturation
  - in the limit of $\bar{u}_\parallel=0$ the NLDR recovers the linear results [A.R.Bell 2004, 2005 MNRAS]

Nonlinear Dispersion Relation

- The most unstable mode is determined by setting $d\gamma_k/dk_\parallel = 0$:

$$k_c/k_{\max} = 1 - \bar{u}_\parallel/u_{cr}$$

$$\gamma_c/\gamma_{\max} = \sqrt{(1-\bar{u}_\parallel/u_{cr})^2 - \epsilon^2} \quad \text{(monotonically decreasing)} \tag{3}$$

- The saturated state is determined by setting $\gamma_c = 0$:
  - the saturation level of the PMF

$$\bar{u}_\parallel/v_A = (u_{cr} - v_A)/v_A \tag{4}$$

  - the frequency-wavenumber spectrum of the saturated state

$$\omega_{kr}/\gamma_{\max} = \frac{v_A}{u_{cr}} + \frac{k_\parallel(u_{cr}-v_A)}{\gamma_{\max}} \tag{5}$$

Mean-Field Equation

- Mean-field equation for the PMF obtained from the spatial average over the momentum equation:

$$\rho_0 \frac{d\bar{u}_\parallel}{dt} = \frac{en_{cr}}{c} \overline{\hat{\mathbf{b}}_0 \cdot (\delta \mathbf{u}_\perp \times \delta \mathbf{B}_\perp)} = \frac{en_{cr}}{c} \sum_{k_\parallel > 0} \overline{\left[\hat{\mathbf{b}}_0 \cdot (\delta \mathbf{u}_{k\perp} \times \delta \mathbf{B}^*_{k\perp}) + c.c.\right]} = \frac{en_{cr}}{cB_0} \sum_{k_\parallel > 0} \frac{2\gamma_k}{k_\parallel} |\delta \mathbf{B}_{k\perp}|^2$$

- An analytical solution for $\bar{u}_\parallel$ is obtained:

$$\rho_0 \bar{u}_\parallel = \frac{en_{cr}}{cB_0} \sum_{k_\parallel > 0} \frac{|\delta \mathbf{B}_{k\perp}|^2}{k_\parallel} \tag{6}$$

  - $|\delta \mathbf{B}_{k\perp}| \propto \exp(\gamma_{\max} t)$, $\bar{u}_\parallel \propto \exp(2\gamma_{\max} t)$
  - the PMF is beat-driven by the non-resonant modes and grows at twice the growth rate of the fastest-growing mode
  - according to the NLDR, the PMF in turn reduces the growth rate of the non-resonant modes and leads to saturation

Energy Increments

- Energy conservation equation obtained from the total energy conservation law adopting mean-field assumptions:

$$\frac{dE^W}{dt} = \frac{d}{dt}\int d\mathbf{r}\underbrace{\left(\frac{\rho|\delta\mathbf{u}|^2}{2} + \frac{|\delta\mathbf{B}|^2}{8\pi} + \frac{p}{\gamma-1}\right)}_{\text{fluctuation energy}} = -\int d\mathbf{r}(J_{cr} - en_{cr}\bar{u}_\parallel)\delta\mathbf{u}_\perp \cdot (\hat{\mathbf{b}}_0 \times \delta\mathbf{B}_\perp)/c$$

$$= \frac{J_{cr} - en_{cr}\bar{u}_\parallel}{cB_0}\sum_{k_\parallel>0}\frac{2\gamma_k}{k_\parallel}|\delta\mathbf{B}_{k\perp}|^2 = \rho_0 u_{cr}\frac{d}{dt}\left(\bar{u}_\parallel - \frac{\bar{u}_\parallel^2}{2u_{cr}}\right)$$

- An analytical solution for the increment of the fluctuation energy ΔE^W:

$$\Delta E^W = \rho_0 u_{cr}\left(\bar{u}_\parallel - \frac{\bar{u}_\parallel^2}{2u_{cr}}\right)$$

- An analytical solution for the increment of the total energy ΔE^{tot}:

$$\Delta E^{tot} = \Delta E^W + \frac{1}{2}\rho_0\bar{u}_\parallel^2 = \rho_0 u_{cr}\bar{u}_\parallel$$

- Substitute in the saturation level of the PMF $\bar{u}_\parallel = u_{cr} - v_A$ gives the final increments after the saturation:

$$\Delta E^W = \frac{1}{2}\rho_0(u_{cr}^2 - v_A^2), \quad \Delta E^{tot} = \rho_0 u_{cr}(u_{cr} - v_A) \tag{7}$$

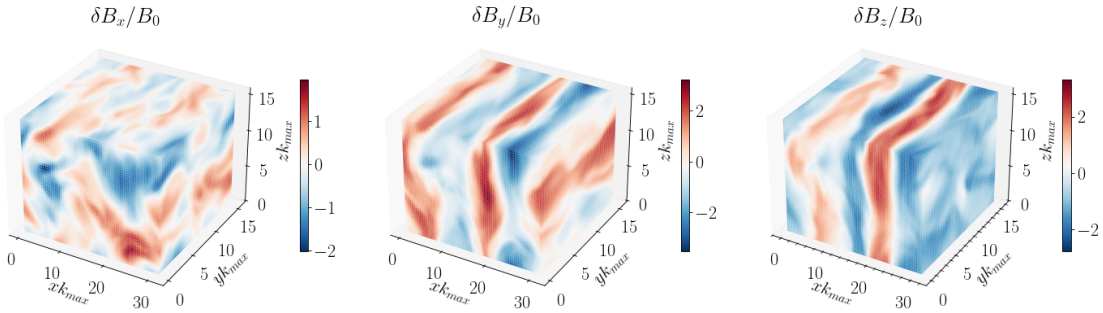
3D Fields

- Simulation setup:
  - periodic boundary conditions
  - initial random-noise fluctuations
  - a fixed, uniform, and constant CR-current and an initial mean magnetic field both in the $\hat{x}$-direction
- The background plasma is set into bulk motion in the parallel direction $\Longrightarrow$ parallel mean flow

3D Fields

Saturated State

- δB_x, δB_y and δB_z are comparable in magnitude
- δB_y and δB_z are correlated in space; stratification in the parallel direction $\Longrightarrow$ incompressible fluctuations

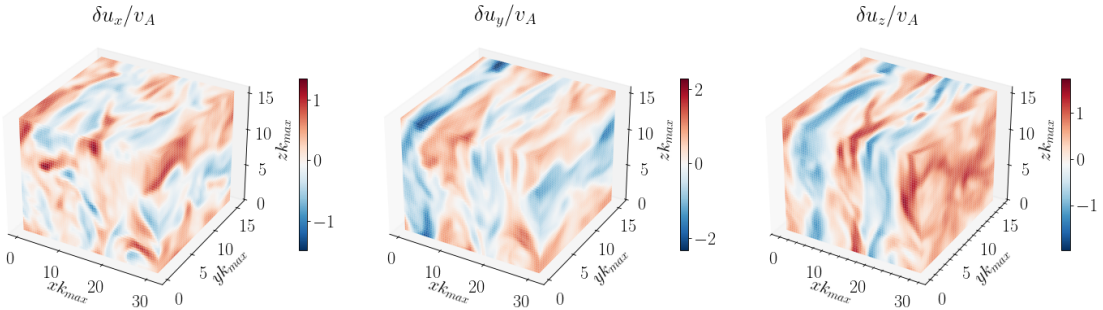


$\delta\mathbf{B}$ obtained from 3D simulation with $\epsilon = 0.3$ at $t\gamma_{\max} = 40$

3D Fields

Saturated State

- δu_x, δu_y and δu_z are comparable in magnitude
- δu_y and δu_z are correlated in space; stratification in the parallel direction $\Longrightarrow$ incompressible fluctuations

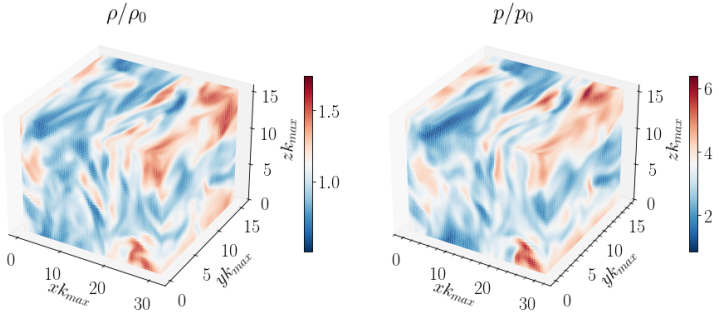


$\delta\mathbf{u}$ obtained from 3D simulation with $\epsilon = 0.3$ at $t\gamma_{\max} = 40$

3D Fields

Saturated State

- ρ and p are correlated in space; chaotic structures $\Longrightarrow$ compressible fluctuations

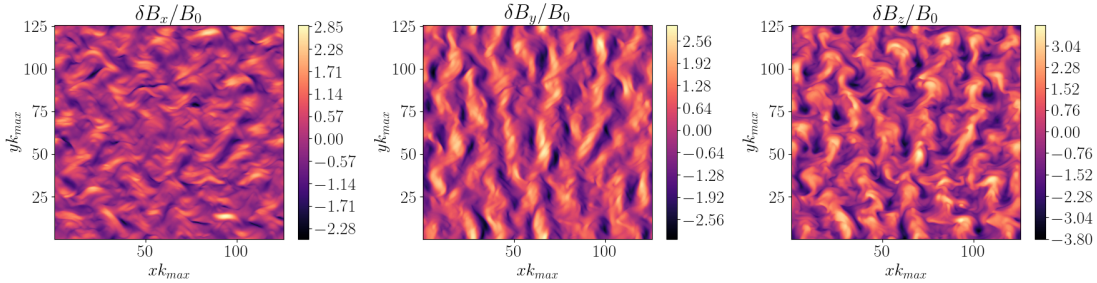


ρ and p obtained from 3D simulation with $\epsilon = 0.3$ at $t\gamma_{\text{max}} = 40$

2D Fields

Saturated State

- δB_x, δB_y and δB_z are comparable in magnitude
- δB_y and δB_z are correlated in space; stratification in the parallel direction

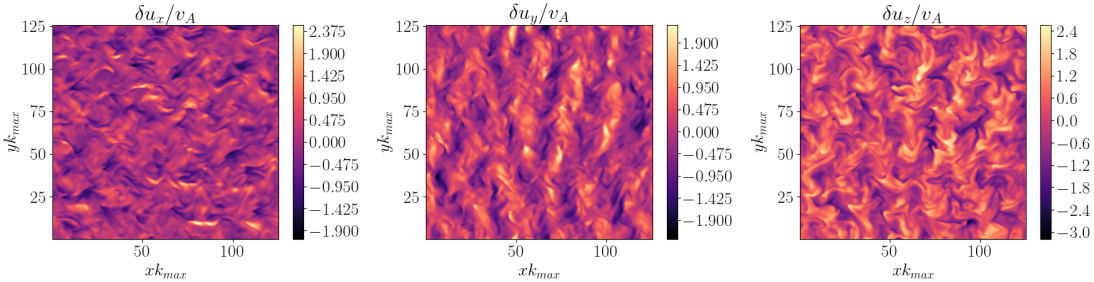


$\delta\mathbf{B}$ obtained from 2D simulation with $\epsilon = 0.3$ at $t\gamma_{\text{max}} = 40$

2D Fields

Saturated State

- δu_x, δu_y and δu_z are comparable in magnitude
- δu_y and δu_z are correlated in space; stratification in the parallel direction

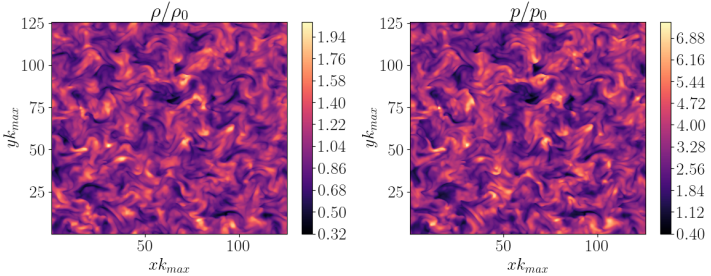


$\delta\mathbf{u}$ obtained from 2D simulation with $\epsilon = 0.3$ at $t\gamma_{\max} = 40$

2D Fields

Saturated State

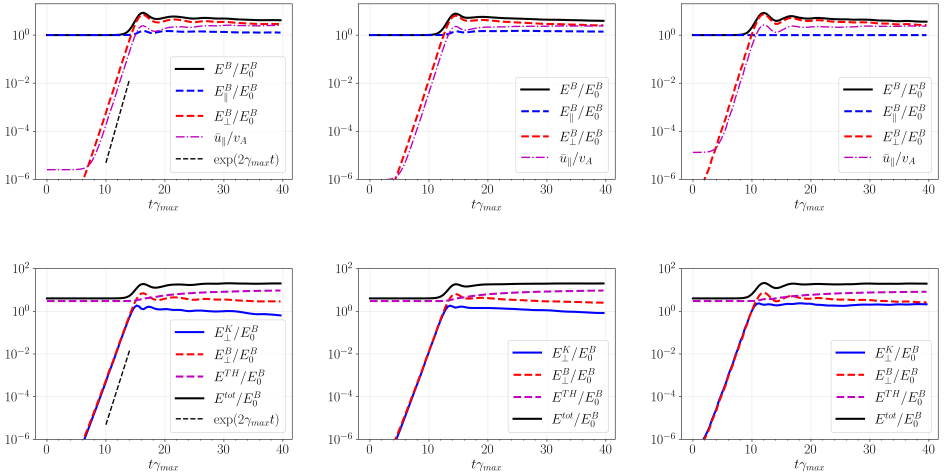
- ρ and p are correlated in space; chaotic structures



ρ and p obtained from 2D simulation with $\epsilon = 0.3$ at $t\gamma_{\rm max} = 40$

Energetics

- Different energy components and the parallel mean flow with $\epsilon = 0.3$, for 3D, 2D, and 1D simulations:

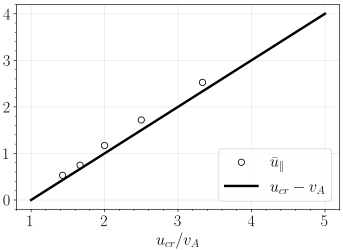


Saturated State

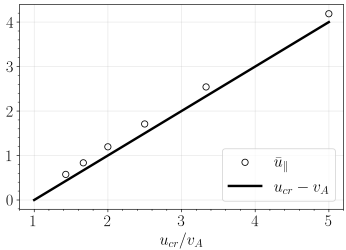
- Saturation levels of the parallel mean flow $\bar{u}_{\parallel}$ with different u_{cr}

$$\bar{u}_{\parallel} = u_{cr} - v_A$$

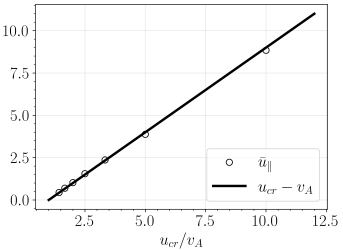
- Quantitative agreement between simulations (circles) and the MFT prediction (solid line)



3D simulations



2D simulations



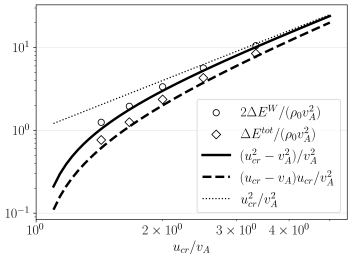
1D simulations

Saturated State

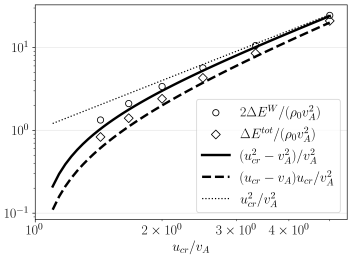
- Increments of fluctuation energy ΔE^W and total energy ΔE^{tot} with different u_{cr}

$$\Delta E^W = \frac{1}{2}\rho_0(u_{cr}^2 - v_A^2), \quad \Delta E^{tot} = \Delta E^W + \frac{1}{2}\rho_0 \bar{u}_\parallel^2 = \rho_0 u_{cr}(u_{cr} - v_A)$$

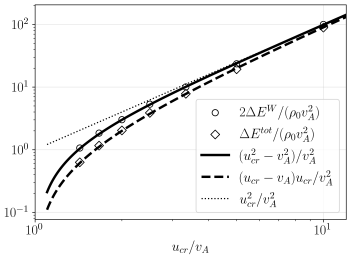
- Quantitative agreement between simulations (circles/diamonds) and the MFT prediction (solid/dashed line)



3D simulations



2D simulations



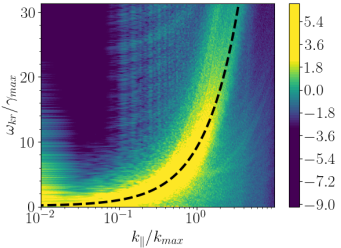
1D simulations

Saturated State

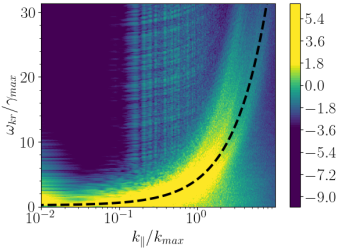
- $(\omega_{kr} - k_\parallel)$ spectra of δB_y given the 2D FFT of 1D simulations during $t\gamma_{\max} \in [20, 100]$

$$\frac{\omega_{kr}}{\gamma_{\max}} = \frac{v_A}{u_{cr}} + \frac{k_\parallel (u_{cr} - v_A)}{\gamma_{\max}}$$

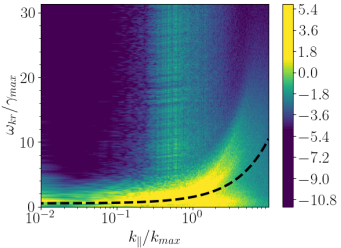
- Qualitative agreement between simulations and the MFT prediction (dashed line)



$\epsilon = 0.1$



$\epsilon = 0.2$



$\epsilon = 0.5$

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Mean-Field Theory

- We consider the self-consistent backreaction on CRs, while assuming CRs to be a cold-beam population throughout the entire evolution
  - the averaged CR-current $\bar{J}_{cr\parallel}(t) = en_{cr}\bar{u}_{cr\parallel}(t)$ evolves in time due to the backreaction on CRs
  - the perturbed CR-current is neglected $\delta\mathbf{J}_{cr\perp} = 0$ under the cold-beam assumption
- Nonlinear dispersion relation remains the same as the fixed CR-current case:

$$(\omega_k - k_\parallel \bar{u}_\parallel)^2 - k_\parallel^2 v_A^2 = \mp \frac{n_{cr}}{n_i}\Omega(\omega_k - k_\parallel \bar{u}_{cr\parallel})$$

- The first saturation criterion remains the same:

$$\bar{u}_{cr\parallel} - \bar{u}_\parallel = v_A \tag{8}$$

where the CR bulk velocity $\bar{u}_{cr\parallel}$ is no longer a constant compared to the fixed CR-current case

Mean-Field Equations

- The mean-field equation for the PMF remains the same as the fixed CR-current case

$$\rho_0 \frac{d\bar{u}_\parallel}{dt} = -en_{cr}\bar{E}_\parallel$$

where the self-induced parallel mean electric field is defined as

$$\bar{E}_\parallel = -\overline{\hat{\mathbf{b}}_0 \cdot (\delta\mathbf{u}_\perp \times \delta\mathbf{B}_\perp)}/c$$

- The averaged Vlasov equation for the averaged CR distribution function $\bar{f}$ due to the effect of $\bar{E}_\parallel$

$$\left(\frac{\partial}{\partial t} + \frac{e}{m}\bar{E}_\parallel \frac{\partial}{\partial v_\parallel}\right)\bar{f} = 0, \qquad \int dv_\parallel \bar{f}(v_\parallel, t) = 0$$

- The momentum equation for the bulk CRs

$$n_{cr} m \frac{d\bar{u}_{cr\parallel}}{dt} = en_{cr}\bar{E}_\parallel$$

- The energy equation for the bulk CRs

$$\frac{d}{dt}\mathcal{K}_{cr} = \bar{J}_{cr\parallel}\bar{E}_\parallel, \quad \mathcal{K}_{cr} \equiv \frac{1}{2}\langle mv^2 \bar{f}\rangle_v$$

Conservation Laws

- Mometum conservation law

$$n_i m \frac{d\bar{u}_\parallel}{dt} = -e n_{cr} \bar{E}_\parallel = -n_{cr} m \frac{d\bar{u}_{cr\parallel}}{dt} \tag{9}$$

- Energy conservation law

$$\frac{dE^W}{dt} + \frac{d}{dt}\frac{\rho_0 \bar{u}_\parallel^2}{2} = -\bar{J}_{cr\parallel}\bar{E}_\parallel = -\frac{d\mathcal{K}_{cr}}{dt} \tag{10}$$

- The self-induced parallel mean electric field $\bar{E}_\parallel$ is responsible for the energy and momentum transfer between the bulk CRs and the bulk plasma
- The second saturation criterion is given by the momentum conservation

$$\bar{u}_\parallel = \frac{n_{cr}}{n_i}(u_{cr\parallel 0} - \bar{u}_{cr\parallel}) \tag{11}$$

Saturated State

- Based on the two saturation criteria

$$\bar{u}_{cr\|} - \bar{u}_{\|} = v_A, \quad \bar{u}_{\|} = \frac{n_{cr}}{n_i}(u_{cr\|0} - \bar{u}_{cr\|})$$

we obtain the saturation levels of the PMF and the bulk CR velocity

$$\bar{u}_{\|} = \frac{(n_{cr}/n_i)(u_{cr\|0} - v_A)}{1 + n_{cr}/n_i}, \quad \bar{u}_{cr\|} = \frac{(n_{cr}/n_i)u_{cr\|0} + v_A}{1 + n_{cr}/n_i} \tag{12}$$

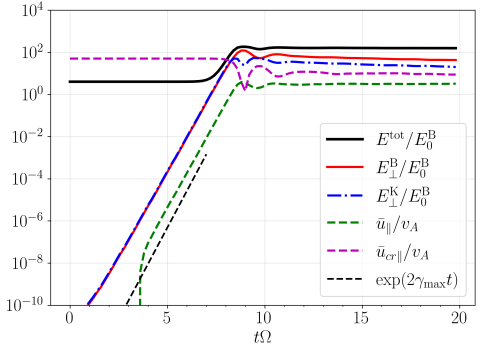
- From the energy equation we derive the increments of the fluctuation energy and the total energy respectively

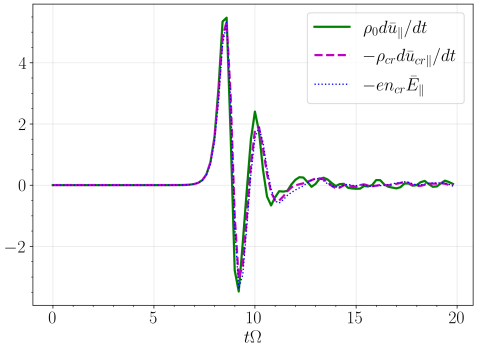
$$\Delta E^W = \frac{1}{2}(\rho_{cr}u_{cr\|0}^2 - \rho_{cr}\bar{u}_{cr\|}^2 - \rho_0\bar{u}_{\|}^2) = \frac{1}{2}\rho_0\frac{n_{cr}}{n_i}\frac{u_{cr\|0}^2 - v_A^2}{1 + n_{cr}/n_i} \tag{13}$$

$$\Delta E^{tot} = \frac{1}{2}\rho_{cr}(u_{cr\|0}^2 - \bar{u}_{cr\|}^2) = \frac{1}{2}\rho_0\frac{n_{cr}}{n_i}\frac{u_{cr\|0}^2 - v_A^2 + 2(n_{cr}/n_i)u_{cr\|0}(u_{cr\|0} - v_A)}{(1 + n_{cr}/n_i)^2} \tag{14}$$

Plasma Dynamics

- 1D simulation with $n_{cr} = 0.08 n_i$ and $u_{cr\|0} = 50 v_A$
  - dynamics of total energy E^{tot}, perpendicular magnetic energy $E_\perp^B$, perpendicular kinetic energy $E_\perp^K$, parallel mean flow $\bar{u}_\|$, and bulk CR velocity $\bar{u}_{cr\|}$
  - momentum conservation law verified



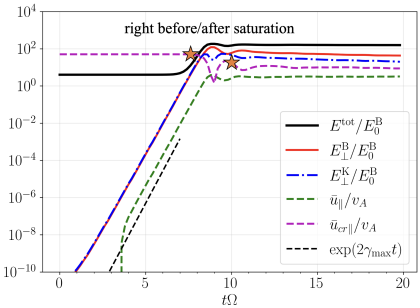


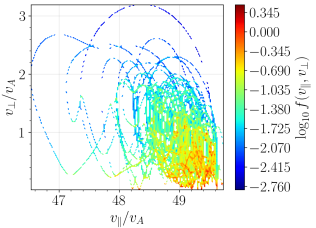
Particle Dynamics

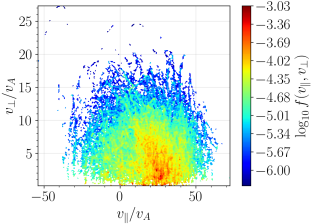
- 1D simulation with $n_{cr} = 0.08 n_i$ and $u_{cr\|0} = 50 v_A$
  - velocity-space distribution function at $t\Omega = 7.5,\ 10$

$$2\pi \int dv_\parallel dv_\perp v_\perp f(v_\parallel, v_\perp) = 1$$

  - weak particle scattering before saturation
  - strong particle diffusion right after saturation





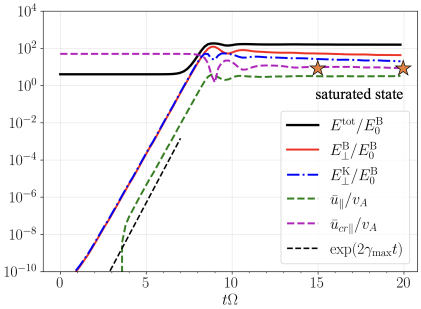


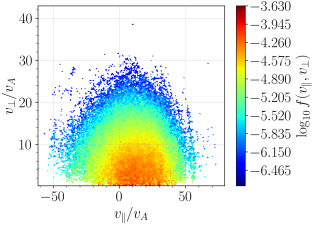
Particle Dynamics

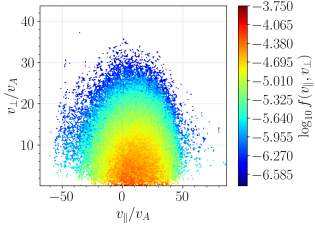
- 1D simulation with $n_{cr} = 0.08 n_i$ and $u_{cr\|0} = 50 v_A$
  - velocity-space distribution function at $t\Omega = 15,\ 20$

$$2\pi \int dv_\parallel dv_\perp v_\perp f(v_\parallel, v_\perp) = 1$$

  - drift bi-Maxwellian distribution after saturation



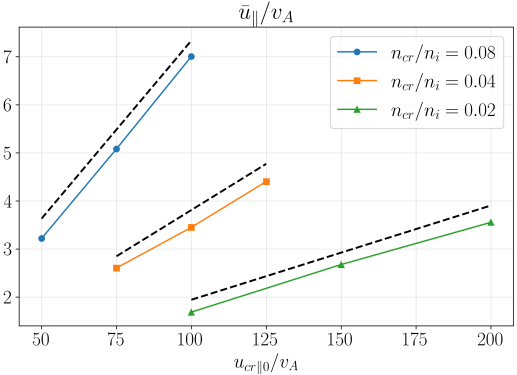


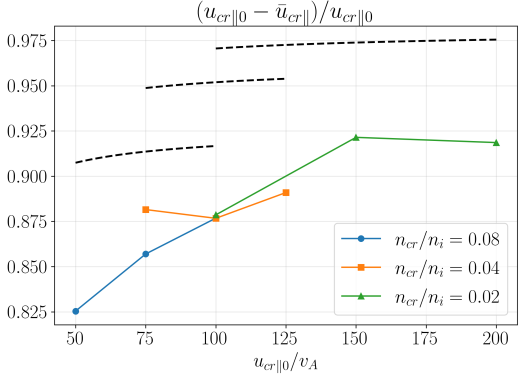


Saturated State

- Saturation levels of the PMF and the bulk CR velocity

$$\bar{u}_{\parallel} = \frac{(n_{cr}/n_i)(u_{cr\parallel 0} - v_A)}{1 + n_{cr}/n_i}, \quad \bar{u}_{cr\parallel} = \frac{(n_{cr}/n_i)u_{cr\parallel 0} + v_A}{1 + n_{cr}/n_i}$$

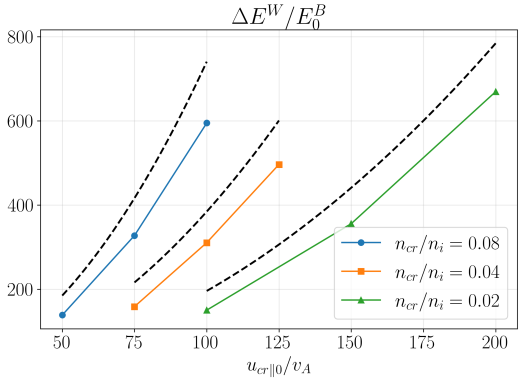


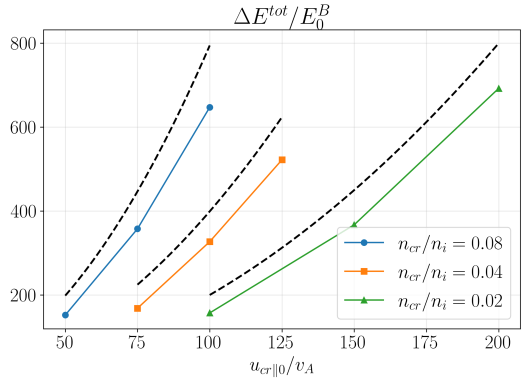


Saturated State

- Increments of the fluctuation energy and the total energy

$$\Delta E^W = \frac{1}{2}\rho_0 \frac{n_{cr}}{n_i} \frac{u_{cr\|0}^2 - v_A^2}{1 + n_{cr}/n_i}, \quad \Delta E^{tot} = \frac{1}{2}\rho_0 \frac{n_{cr}}{n_i} \frac{u_{cr\|0}^2 - v_A^2 + 2(n_{cr}/n_i)u_{cr\|0}(u_{cr\|0} - v_A)}{(1 + n_{cr}/n_i)^2}$$





Outline

1. Background & Motivation

2. Mean-Field Theory for the Fixed CR-Current Case
2.1 Analytical Theory
2.2 Multi-dimensional MHD Simulations

3. Mean-Field Theory for the Self-Consistent CR Case
3.1 Analytical Theory
3.2 One-dimensional MHD-PIC Simulations

4. Conclusion, Discussion & Extension

Conclusion

- A shift of physical picture: wave-particle scattering (RSI) $\Longrightarrow$ mean-field dynamics (NRSI); the saturation largely depends on n_{cr}/n_i and $u_{cr\|0}$
- For the fixed CR-current case: the PMF dynamics dominates the saturation
  - saturation criterion given by the NLDR: $\bar{u}_\| = u_{cr} - v_A$
  - saturation levels and energy increments predicted by the MFT are quantitatively verified by multi-dimensional MHD simulations
- For the self-consistent CR case: the mean-field dynamics of both the PMF and the bulk CRs dominates the saturation
  - the first saturation criterion given by the NLDR: $\bar{u}_{cr\|} - \bar{u}_\| = v_A$
  - the second saturation criterion given by the momentum conservation: $\bar{u}_\| = (n_{cr}/n_i)(u_{cr\|0} - \bar{u}_{cr\|})$
  - saturation levels and energy increments predicted by the MFT are verified by 1D MHD-PIC simulations
- Simple physical explanations for the two criteria:
  - the effective streaming speed of CRs vanishes in the plasma co-moving frame
  - the bulk momemtum of the CRs and the plasma is conserved

Discussion

- Remark 1: The effect of wave-particle interaction due to the perturbed CR-current $\delta\mathbf{J}_{cr\perp}$:
  - nonlinear dispersion relation with wave-particle interaction included [E.Amato & P.Blasi 2009 MNRAS]

$$(\omega_k - k_\parallel \bar{u}_\parallel)^2 - k_\parallel^2 v_A^2 = \mp \frac{n_{cr}}{n_i}\Omega(\omega_k - k_\parallel \bar{u}_{cr\parallel})(1 + \underbrace{I_1 + iI_2}_{\delta\mathbf{J}_{cr\perp}}) \tag{15}$$

  with the non-resonant effect $0 < 1 + I_1 < 1$ and the resonant effect $I_2 < 0$, one may calculate the saturation criterion assuming I_1, I_2 are constants

$$\bar{u}_{cr\parallel} - \bar{u}_\parallel = v_A\sqrt{1 + I_2^2/(1+I_1)^2} > v_A \tag{16}$$

  - by including $\delta\mathbf{J}_{cr\perp}$, one may predict a smaller value of $\bar{u}_\parallel$, a larger value of $\bar{u}_{cr\parallel}$, and smaller values of ΔE^W and ΔE^{tot}, therefore a better agreement with simulation results
- Remark 2: Direct mode-couplings among fluctuations and damping effects neglected here should play a role in redistributing wave energy among different parts of the total energy, in which case additional closures or stochastic models are required

Extension

- Extension 1: Multi-dimensional simulations for the self-consistent CR case with different shapes of the CR distribution, e.g. drift Maxwellian and drift power-law
- Extension 2: An analytical theory combining the present mean-field theory and the quasi-linear theory should capture the detailed particle dynamics quantitatively
- Extension 3: A unified theory for the saturation of the cosmic-ray self-generated streaming instability is possible

regime	resonant streaming instability	non-resonant streaming instability
type of instability	dissipative (kinetic)	reactive (fluid)
source of instability	resonant wave-particle interaction	non-resonant CR-current drive
dominant dynamics	particle diffusion & wave damping	bulk plasma motion & bulk CR motion
definition of problem	near-equilibrium relaxation	out-of-equilibrium process
theoretical description	quasi-linear theory	mean-field theory
saturation mechanism	balance between instability and damping	mean-field dynamics
turbulent level	weak turbulence	strong turbulence